ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE G · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice g.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches, tables and circuits.

QUESTION 1

1.1 Any FIVE sensible, measurable items ✓✓✓✓✓, e.g. drills at least 60 shelves per hour; hole position accurate to ± 0,5 mm; hole depth 12 mm; works on 6 bar compressed air and 230 V; shelves up to 900 mm × 400 mm × 25 mm; cost below R45 000; two-hand start and emergency stop; noise below 85 dB; fits in a floor space of 2 m × 1,5 m. (5)

1.2 Measurable items can be tested and checked when the machine is finished ✓, so there is no argument about whether the design meets the client's needs ✓. Example from 1.1 ✓, e.g. "at least 60 shelves per hour" can be timed. (3)

1.3 Any TWO ✓✓: interviews / meetings; questionnaires; watching the operators at work (observation); studying the existing drawings, records and complaints; site visit and measurements. (2)

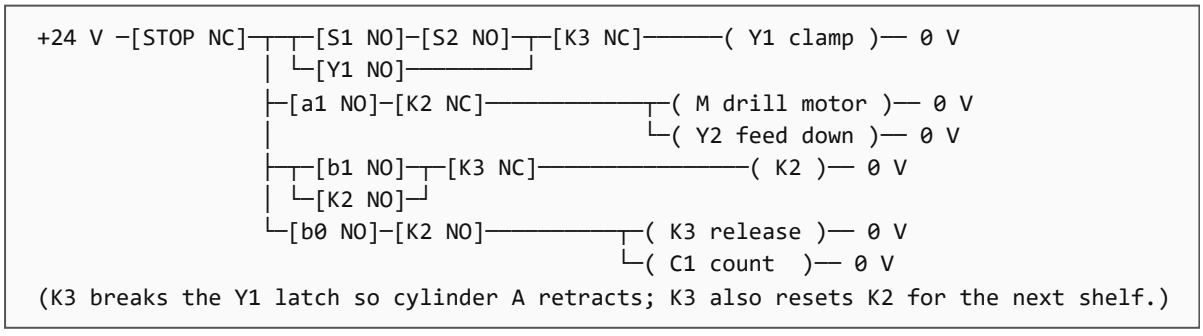
[10]

QUESTION 2

2.1 Two horizontal lines (cylinders A and B) with steps 1–5 marked ✓; A rises (extends) between steps 1 and 2 ✓ and stays out ✓; B extends between steps 2 and 3 ✓ and retracts between steps 3 and 4 ✓; A retracts between steps 4 and 5 ✓ (6)

2.2 Any FIVE components with their functions ✓✓✓✓✓, e.g. compressor / air supply – provides compressed air; service unit (filter, regulator, lubricator) – cleans the air and sets the pressure; 5/2-way solenoid valves – direct the air to extend or retract each cylinder; double-acting cylinders – clamp the shelf and feed the drill; flow control valves – set the speed of the drill feed; reed switches / limit switches – signal the cylinder positions; silencers – reduce exhaust noise. (5)

2.3 S1 and S2 in series ✓ with a Y1 holding contact in parallel ✓ through STOP (NC) ✓ and an NC contact of the release step ✓ driving Y1 ✓; a1 driving M and Y2 through an NC K2 contact ✓; b1 latching K2 ✓; b0 and K2 in series releasing the clamp (breaking the Y1 latch) ✓ and counting C1 ✓; correct symbols and labels ✓, e.g. (10)



2.4 Any TWO methods, each with an advantage ✓✓ ✓✓, e.g. foot or flange mountings bolted with nuts, bolts and lock washers – strong and easy to remove for maintenance; trunnion / clevis mountings – allow the cylinder to pivot; welded brackets – very strong and rigid; T-slot profiles with T-nuts – positions are easy to adjust. (4)

[25]

QUESTION 3

3.1 Suitable headings ✓✓✓ (item number, description / part number, quantity, unit cost, total cost, supplier); at least FIVE sensible items ✓✓✓✓ (e.g. two double-acting cylinders, two 5/2 solenoid valves, service unit, drill motor and chuck, reed switches, PLC or relays, steel frame, push buttons and E-stop); grand total ✓ (8)

3.2 The critical path is the longest chain of activities that depend on each other ✓, which sets the shortest possible time for the whole project ✓. A delay on any activity on the critical path delays the whole project ✓, so the project manager must watch those activities closely and plan resources for them ✓. (4)

3.3 Week scale 1–8 along the top ✓; SIX sensible activities listed ✓✓ (e.g. design, ordering, building the frame, pneumatics, electrical and PLC, testing and commissioning); bars of sensible length ✓✓; correct order of activities ✓; overlapping activities shown ✓; neat, with a title and key ✓ (8)

3.4 A risk register is a list of the things that could go wrong in the project, how likely and serious each one is, and how it will be controlled ✓✓. TWO risks with controls ✓✓ ✓✓, e.g. late delivery of the valves – order early, choose a second supplier; costs higher than the budget – get three quotes, keep a contingency; injury during assembly – safety talk and PPE; drill position wrong – build and test a prototype. (6)

3.5 A prototype shows whether the design works in practice ✓ and finds mistakes early ✓, when they are cheaper and easier to fix ✓; the client and operators can also try it and suggest changes ✓. (4)

[30]

QUESTION 4

4.1 Cold commissioning: checks done without the machine working under power and air, e.g. wiring, connections, pipe routing, I/O checks and the program in simulation ✓✓. Hot commissioning: the machine is switched on with air and power and run with real shelves to check the full sequence and performance ✓✓. (4)

4.2 Any FIVE sensible checks ✓✓✓✓✓, e.g. emergency stop stops everything; both start buttons needed; air pressure set at 6 bar and no leaks; clamp holds the shelf firmly; hole position and depth correct; counter counts correctly; guards fitted; required output per hour reached. (5)

4.3 Any THREE ✓✓✓: reed switch b1 faulty, loose or set in the wrong position; broken wire or loose terminal on b1 or the K2 circuit; solenoid valve Y2 sticking (spool jammed); K2 relay faulty; program / wiring error; low air pressure. (3)

4.4 They protect the operator's hands and body ✓; if they do not work, any other test could cause injury ✓, so the station must not be run until they work correctly ✓. (3)

[15]

QUESTION 5

5.1 A lubrication chart shows which points must be lubricated, with which lubricant and how often ✓✓, so that no point is missed. Any TWO parts ✓✓: drill spindle bearings / quill; guide rails or slides of the drill feed; cylinder rod pivots / clevis pins; service unit lubricator (oil level). (4)

5.2 With the system under pressure ✓, listen for hissing and spray soapy water (leak-detection fluid) on the fittings, hoses and valves ✓; bubbles show a leak, which must be repaired after the system is isolated and the pressure released ✓. (3)

5.3.1 $\text{MTBF} = \text{operating time} \div \text{number of failures} = 1\,200 \div 4 \checkmark = 300 \text{ h} \checkmark$ (2)

5.3.2 $\text{MTTR} = \text{repair time} \div \text{number of failures} = 24 \div 4 \checkmark = 6 \text{ h} \checkmark$ (2)

5.3.3 $\text{Availability} = \text{MTBF} \div (\text{MTBF} + \text{MTTR}) \times 100\% = 300 \div 306 \times 100\% \checkmark$ (2)
$= 98{,}04\% \checkmark$

5.4 Any FOUR steps in order ✓✓✓✓: tell the operators; stop the machine; switch off and isolate the electrical supply; close the air supply valve and release the stored air pressure; fit your own lock and a tag (name, date, reason) to each isolator; test that the machine cannot start (try the start buttons); remove the lock only when the work is finished. (4)

5.5 Any THREE ✓✓✓: "Danger – moving parts / keep hands clear"; "Eye protection must be worn"; "Hearing protection must be worn"; "Compressed air – isolate before maintenance"; "Emergency stop" label; "Only trained operators". (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE